PROBLEM CL15

Mixing Elbow: Exit Velocity and Temperature, Force and Torque

Original statement · CL15.pdf

Two inlets | Mixing | Energy | Angular momentum

OFFICIAL PROBLEM STATEMENT

Consider the mixing elbow shown schematically in the figure of CL15.pdf. Cold water flows through the main pipe and mixes with warmer water that enters through a smaller inlet located at the elbow. The liquid properties can be considered independent of the temperature and gravity effects can be neglected. All pipe walls are adiabatic. The known pipe dimensions, fluid properties and boundary conditions are given in the figure (density ρ, viscosity μ, conductivity k, specific heat c; main inlet v_1, p_1, T_1, diameter D_1; secondary inlet v_2, p_2, T_2, diameter D_2; exit at p_a). Assume that the two streams mix completely and that the velocity and temperature are uniform at the exit section.

- Find the velocity v_s and temperature T_s of the mixed stream emerging from the pipe at ambient pressure.
- Obtain the net force $\bar{F}$ exerted on the mixing elbow by the water flowing inside and the ambient air.
- Obtain the torque $\bar{M}_1$ about the point $\bar{x}_1$ located at the center of the larger inlet.
- Compute the above results numerically, given the following data: $v_1 = 0.4$ m/s; $v_2 = 1.2$ m/s; $\rho = 1000$ kg/m^3; $T_1 = 20^\circ$C; $T_2 = 40^\circ$C; $\mu = 8 \times 10^{-4}$ Pa s; $p_1 - p_a = 50$ Pa; $p_2 - p_a = 200$ Pa; $k = 0.677$ W/(m s); $D_1 = 10$ cm; $D_2 = 2.5$ cm; $c = 4216$ J/(kg K); $L_1 = 20$ cm; $L_2 = 7$ cm.

Source: CL15.pdf (pages 1-2, exam of Dec 18, 2012, with solution). Notes.pdf, Chapter 3: integral mass balance Eq. 3.11, integral momentum balance Eq. 3.36, integral angular momentum balance Eq. 3.38, integral total energy balance Eq. 3.46.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady, incompressible liquid with constant properties, $e = e_0 + cT$; neglect $\bar{g}$; adiabatic walls; uniform properties at the three sections; complete mixing.
- Geometry (figure): main inlet along $+\bar{e}_x$ (section $A_1 = \pi D_1^2/4$), secondary inlet along $+\bar{e}_y$ ($A_2 = \pi D_2^2/4$), exit along $+\bar{e}_y$ (section A_1) at p_a. The secondary inlet and the exit share the same vertical axis, at a horizontal distance $L_1 + D_1 + D_1/2 = L_1 + \frac{3}{2}D_1$ from $\bar{x}_1$ (bend of inner radius D_1; reading of the figure, as in the official solution).
- Data: as listed. Unknowns: $[v_s,\ T_s,\ \bar{F},\ \bar{M}_1]$. μ, k and L_2 are not needed.

Symbol	Meaning	SI unit
v_1, v_2, v_s	inlet and exit velocities	m/s
T_1, T_2, T_s	temperatures	°C
$p_1 - p_a, p_2 - p_a$	inlet gauge pressures	Pa
D_1, D_2, L_1	diameters and length	m
$\bar{F}, \bar{M}_1$	force and torque on the elbow	N, N·m

PHASE 2

Frames, Control Volume and Surfaces

- Frame fixed to the elbow: $\bar{e}_x$ horizontal (main inlet flow), $\bar{e}_y$ vertical upwards, $\bar{e}_z$ towards the reader; moments about $\bar{x}_1$.
- Surfaces (fixed CV, as in the official solution): Σ_1 (main inlet, $\bar{n} = -\bar{e}_x$), Σ_2 (secondary inlet, $\bar{n} = -\bar{e}_y$), Σ_{out} (exit, $\bar{n} = +\bar{e}_y$), Σ_{wall}.

$$\boxed{\bar{F} = -\int_{\Sigma_{wall}} \left[-(p - p_a)\,\bar{n} + \bar{\bar{\tau}}' \cdot \bar{n}\right] d\sigma}$$

$$\boxed{\bar{M}_1 = -\int_{\Sigma_{wall}} (\bar{x} - \bar{x}_1) \wedge \left[-(p - p_a)\,\bar{n} + \bar{\bar{\tau}}' \cdot \bar{n}\right] d\sigma}$$

PHASE 3

Balances, Surface by Surface

Mass (Notes Eq. 3.11)

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c} \rho\, dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c} \rho\,(\bar{v} - \underbrace{\cancel{\bar{v}_c}}_{0}) \cdot \bar{n}\, d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\, \bar{v} \cdot \bar{n}\, d\sigma$
Σ_1	$-\rho v_1 A_1$
Σ_2	$-\rho v_2 A_2$
Σ_{out}	$+\rho v_s A_1$
Σ_{wall}	0

$$v_1 A_1 + v_2 A_2 = v_s A_1 \implies \boxed{v_s = v_1 + v_2 \left(\frac{D_2}{D_1}\right)^2} \qquad (1)$$

Total energy (Notes Eq. 3.46)

$$\underbrace{\cancel{\frac{d}{dt}\int_{V_c} \rho\left(e + \frac{v^2}{2}\right) dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c} \rho\left(e + \frac{v^2}{2}\right)\left[(\bar{v} - \underbrace{\cancel{\bar{v}_c}}_{0}) \cdot \bar{n}\right] d\sigma = -\int_{\Sigma_c} p\,\bar{v} \cdot \bar{n}\, d\sigma + \int_{\Sigma_c} \bar{v} \cdot (\bar{\bar{\tau}}' \cdot \bar{n})\, d\sigma + \underbrace{\cancel{\int_{V_c} \rho \bar{g} \cdot \bar{v}\, dV}}_{\text{neglect } \bar{g}} - \int_{\Sigma_c} \bar{q} \cdot \bar{n}\, d\sigma + \underbrace{\cancel{\int_{V_c} Q_H\, dV}}_{0}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho \left(e + \frac{v^2}{2}\right)(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int p\,\bar{v}\cdot\bar{n}\,d\sigma$	$\int \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$	$-\int \bar{q}\cdot\bar{n}\,d\sigma$
Σ_1	$-\rho\left(e_1+\frac{v_1^2}{2}\right)v_1A_1$	$+p_1v_1A_1$	0	0
Σ_2	$-\rho\left(e_2+\frac{v_2^2}{2}\right)v_2A_2$	$+p_2v_2A_2$	0	0
Σ_{out}	$+\rho\left(e_s+\frac{v_s^2}{2}\right)v_sA_1$	$-p_av_sA_1$	0	0
Σ_{wall}	0	0	0	0 (adiabatic)

With $e = e_0 + cT$, dividing by $\rho c\, v_s A_1$ (e_0 cancels by (1)) and writing p_a as $p_a(v_1A_1 + v_2A_2)/(v_sA_1)$:

$$\boxed{T_s = \left(T_1 + \frac{p_1 - p_a}{\rho c} + \frac{v_1^2}{2c}\right)\frac{v_1}{v_s} + \left(T_2 + \frac{p_2 - p_a}{\rho c} + \frac{v_2^2}{2c}\right)\frac{v_2A_2}{v_sA_1} - \frac{v_s^2}{2c}} \quad (2)$$

Linear momentum (Notes Eq. 3.36)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c}\rho\bar{v}\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c}\rho\bar{v}\,[(\bar{v} - \underbrace{\bar{v}_c}_{0})\cdot\bar{n}]\,d\sigma = -\int_{\Sigma_c}(p - p_a)\,\bar{n}\,d\sigma + \int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma + \underbrace{\cancel{\int_{V_c}\rho\bar{g}\,dV}}_{\text{neglect }\bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\bar{v}\,(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int (p-p_a)\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
Σ_1	$-\rho v_1^2 A_1\,\bar{e}_x$	$(p_1 - p_a)A_1\,\bar{e}_x$	0
Σ_2	$-\rho v_2^2 A_2\,\bar{e}_y$	$(p_2 - p_a)A_2\,\bar{e}_y$	0
Σ_{out}	$+\rho v_s^2 A_1\,\bar{e}_y$	0 ($p = p_a$)	0
Σ_{wall}	0	$-\int (p-p_a)\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$

Boxed terms: $-\bar{F}$.

$$\boxed{\bar{F} = \left(\rho v_1^2 + p_1 - p_a\right)A_1\,\bar{e}_x + \left[\left(\rho v_2^2 + p_2 - p_a\right)A_2 - \rho v_s^2 A_1\right]\bar{e}_y} \quad (3)$$

Angular momentum about $\bar{x}_1$ (Notes Eq. 3.38)

$$\underbrace{\frac{d}{dt}\int_{V_c}\rho\,(\bar{x}-\bar{x}_1)\wedge\bar{v}\,dV}_{0}+\int_{\Sigma_c}\rho\,[(\bar{x}-\bar{x}_1)\wedge\bar{v}](\bar{v}\cdot\bar{n})\,d\sigma=-\int_{\Sigma_c}(\bar{x}-\bar{x}_1)\wedge(p-p_a)\bar{n}\,d\sigma+\int_{\Sigma_c}(\bar{x}-\bar{x}_1)\wedge(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int\rho\,[(\bar{x}-\bar{x}_1)\wedge\bar{v}](\bar{v}\cdot\bar{n})\,d\sigma$	$-\int(\bar{x}-\bar{x}_1)\wedge(p-p_a)\bar{n}\,d\sigma$	$\int(\bar{x}-\bar{x}_1)\wedge(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$
Σ_1	0 (centred at $\bar{x}_1$)	0	0
Σ_2	$-\rho v_2^2 A_2\,d\,\bar{e}_z$	$(p_2-p_a)A_2\,d\,\bar{e}_z$	0
Σ_{out}	$+\rho v_s^2 A_1\,d\,\bar{e}_z$	0	0
Σ_{wall}	0	$-\int(\bar{x}-\bar{x}_1)\wedge(p-p_a)\bar{n}\,d\sigma$	$\int(\bar{x}-\bar{x}_1)\wedge(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$

Boxed terms: $-\bar{M}_1$. With $d=L_1+\frac{3}{2}D_1$ the horizontal distance to the common vertical axis, $(\bar{x}-\bar{x}_1)\wedge\bar{e}_y=d\,\bar{e}_x\wedge\bar{e}_y=d\,\bar{e}_z$ for the centres of Σ_2 and Σ_{out} (the vertical offsets are parallel to $\bar{e}_y$ and drop out).

$$\boxed{\bar{M}_1=\left[\left(\rho v_2^2+p_2-p_a\right)A_2-\rho v_s^2 A_1\right]\left(L_1+\frac{3}{2}D_1\right)\bar{e}_z}\qquad(4)$$

Numerical values (part 4)

Quantity	**Value**
v_s	0.475 m/s
T_s	23.16 °C
$\bar{F}$	$1.65\,\bar{e}_x-0.967\,\bar{e}_y$ N
$\bar{M}_1$	$-0.338\,\bar{e}_z$ N·m

PHASE 4

Interpretation, Limits and Dimensional Check

- (2) is a mass-weighted average of the inlet temperatures, corrected by the tiny flow-work and kinetic-energy terms ($\Delta p/(\rho c)\sim 10^{-5}$ K here): $T_s\approx(v_1A_1T_1+v_2A_2T_2)/(v_sA_1)$.

- $F_y < 0$: the exit momentum flux exceeds the momentum and gauge pressure brought in by the small inlet, so the elbow is pulled downwards; $\bar{M}_1$ follows from the same vertical imbalance acting at the arm d.
- Dimensions (SI): $[(p - p_a)/(\rho c)] = \mathrm{K}$, $[v^2/c] = \mathrm{K}$; $[\rho v^2 A] = \mathrm{N}$; $[\rho v^2 A d] = \mathrm{N \cdot m}$.
- Agreement with the official solution: CL15.pdf obtains (1) to (4) and $v_s = 0.475$ m/s, $T_s = 23^\circ$C, $\bar{F} = 1.65\,\bar{e}_x - 0.967\,\bar{e}_y$ N, $\bar{M}_1 = -0.338\,\bar{e}_z$ N·m, the same values.